



2 Real Analysis 2023

Tutorial 7

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Cauchy sequences

Problem 25

- (a) Show that if $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence, then $\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = 0$.
- (b) Use the definition of a Cauchy sequence to show that the sequence of partial sums of the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is not a Cauchy sequence.
- (c) Give an example of a sequence $(a_n)_{n=1}^{\infty}$ with $\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = 0$ which is not a Cauchy sequence.

Solution.

- (a) Suppose that $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence. Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$|a_m - a_n| < \varepsilon \quad \text{for all } n, m \geq N.$$

If $n \geq N$, then also $n+1 \geq N$, and thus

$$|a_{n+1} - a_n| < \varepsilon \quad \text{for all } n \geq N.$$

This is exactly the definition of the limit $\lim_{n \rightarrow \infty} (a_{n+1} - a_n) = 0$.

- (b) Let $s_n = \sum_{k=1}^n \frac{1}{k}$. Then for $m \in \mathbb{N}$

$$\begin{aligned} s_{2^{m+1}} - s_{2^m} &= \frac{1}{2^m + 1} + \frac{1}{2^m + 2} + \cdots + \frac{1}{2^{m+1}} \\ &\geq \underbrace{\frac{1}{2^{m+1}} + \frac{1}{2^{m+1}} + \cdots + \frac{1}{2^{m+1}}}_{2^m \text{ terms}} = \frac{2^m}{2^{m+1}} = \frac{1}{2}. \end{aligned}$$

Therefore, for any $N \in \mathbb{N}$ we can find $m \in \mathbb{N}$ with $2^m, 2^{m+1} \geq N$ and $|s_{2^{m+1}} - s_{2^m}| \geq \frac{1}{2}$. Hence, $(s_n)_{n=1}^{\infty}$ is not a Cauchy sequence.

- (c) The sequence $(s_n)_{n=1}^{\infty}$ of the partial sums $s_n = \sum_{k=1}^n \frac{1}{k}$ gives an example of a sequence with desired properties. Indeed, by above $(s_n)_{n=1}^{\infty}$ is not a Cauchy sequence, but

$$\lim_{n \rightarrow \infty} (s_{n+1} - s_n) = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0.$$

Absolute and conditional convergence

Problem 26 For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent.

- (a) $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{2n+1}}$, (b) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+1)}$, (c) $\sum_{n=2}^{\infty} (-1)^n \frac{n}{\ln n}$,
 (d) $\sum_{n=0}^{\infty} (-1)^n \frac{5}{n^3+1}$, (e) $\sum_{n=0}^{\infty} \frac{(-5)^n}{n!}$, (f) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^2+3}{(2n-5)^2}$,
 (g) $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt[3]{n}}{n+1}$, (h) $\sum_{n=1}^{\infty} \frac{\cos\left(n\frac{\pi}{6}\right)}{n^2}$, (i) $\sum_{n=1}^{\infty} \frac{n^n}{(-3)^n}$.

Solution.

- (a) The series $\sum_{n=0}^{\infty} \frac{1}{\sqrt{2n+1}}$ diverges, because $\frac{1}{\sqrt{2n+1}} \geq \frac{1}{\sqrt{2n+2}} = \frac{1}{\sqrt{2}} \frac{1}{\sqrt{n+1}}$ for $n \geq 1$, and the p -series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ diverges. On the other hand, the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{2n+1}}$ converges by the Alternating Series Test, because the sequence $\left(\frac{1}{\sqrt{2n+1}}\right)$ is decreasing and converges to zero. We conclude that $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{2n+1}}$ is conditionally convergent.
- (b) The series $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$ diverges, because $\frac{1}{\ln(n+1)} \geq \frac{1}{n}$, and the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. On the other hand, the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+1)}$ converges by the Alternating Series Test, because the sequence $\left(\frac{1}{\ln(n+1)}\right)$ is decreasing and converges to zero. We conclude that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\ln(n+1)}$ is conditionally convergent.
- (c) The series $\sum_{n=2}^{\infty} (-1)^n \frac{n}{\ln n}$ diverges, because the terms $(-1)^n \frac{n}{\ln n}$ do not tend to zero.
- (d) The series $\sum_{n=0}^{\infty} \frac{5}{n^3+1}$ converges, because $\frac{5}{n^3+1} \leq \frac{5}{n^3}$ for $n \geq 1$, and the p -series $\sum_{n=1}^{\infty} \frac{5}{n^3}$ converges. Thus, the series $\sum_{n=0}^{\infty} (-1)^n \frac{5}{n^3+1}$ is absolutely convergent.
- (e) The series $\sum_{n=0}^{\infty} \frac{5^n}{n!}$ converges by the Ratio Test:

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{5^{n+1}}{(n+1)!} \frac{n!}{5^n} = \lim_{n \rightarrow \infty} \frac{5}{n+1} = 0 < 1.$$

We conclude that $\sum_{n=0}^{\infty} \frac{(-5)^n}{n!}$ is absolutely convergent.

- (f) The series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^2+3}{(2n-5)^2}$ diverges, because

$$\lim_{n \rightarrow \infty} \frac{n^2+3}{(2n-5)^2} = \lim_{n \rightarrow \infty} \frac{n^2+3}{4n^2-20n+25} = \lim_{n \rightarrow \infty} \frac{1+\frac{3}{n^2}}{4-\frac{20}{n}+\frac{25}{n^2}} = \frac{1}{4} \neq 0,$$

and thus the terms do not tend to zero.

- (g) The series $\sum_{n=1}^{\infty} \frac{\sqrt[3]{n}}{n+1}$ diverges, because $\frac{\sqrt[3]{n}}{n+1} \geq \frac{\sqrt[3]{n}}{n+n} = \frac{1}{2} \frac{1}{n^{2/3}}$, and the p -series $\sum_{n=1}^{\infty} \frac{1}{2} \frac{1}{n^{2/3}}$ diverges.

The series $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt[3]{n}}{n+1}$ is an alternating series. We have $\lim_{n \rightarrow \infty} \frac{\sqrt[3]{n}}{n+1} = 0$. To apply the Alternating Series Test, we have to show that the sequence $\left(\frac{\sqrt[3]{n}}{n+1}\right)$ is decreasing. To see this, we consider the function

$$\varphi(x) = \frac{x^{1/3}}{x+1}, \quad x \geq 1.$$

Its derivative

$$\varphi'(x) = \frac{1-2x}{3x^{2/3}(x+1)^2}$$

is negative for $x \geq 0$, and thus φ is decreasing. In particular, $\varphi(n) \geq \varphi(n+1)$; this is exactly the statement that the sequence $\left(\frac{\sqrt[3]{n}}{n+1}\right)$ is decreasing.

We conclude that $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt[3]{n}}{n+1}$ converges by the Alternating Series Test, and therefore is conditionally convergent.

- (h) We have $0 \leq \left| \frac{\cos(n\frac{\pi}{6})}{n^2} \right| \leq \frac{1}{n^2}$. Since $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, the series $\sum_{n=1}^{\infty} \left| \frac{\cos(n\frac{\pi}{6})}{n^2} \right|$ converges by the Comparison Test. Thus, $\sum_{n=1}^{\infty} \frac{\cos(n\frac{\pi}{6})}{n^2}$ converges absolutely.

Remark: Notice that $\sum_{n=1}^{\infty} \frac{\cos(n\frac{\pi}{6})}{n^2}$ is not an alternating series.

- (i) The series $\sum_{n=1}^{\infty} \frac{n^n}{(-3)^n}$ diverges, because the terms $\frac{n^n}{(-3)^n}$ do not tend to zero.

Alternating series

Problem 27 For each of the following series, determine how many terms of the series are required to approximate the sum of the series to four decimal places. The latter means that the difference between the sum and the partial sum must be $< \frac{1}{2} \cdot 10^{-4}$.

$$(a) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}, \quad (b) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^n}.$$

Solution.

If $s = \sum_{n=1}^{\infty} (-1)^{n-1} a_n$ and $s_N = \sum_{n=1}^N (-1)^{n-1} a_n$, where $a_n = \frac{1}{n^2}$ or $a_n = \frac{1}{n^n}$, then

$$|s - s_N| \leq a_{N+1}.$$

We want to have $a_{N+1} < \frac{1}{2} \cdot 10^{-4} = \frac{1}{20000}$.

- (a) For the first series, this is the inequality $\frac{1}{(N+1)^2} < \frac{1}{20000}$, i.e. $(N+1)^2 > 20000$. We must have $N > \sqrt{20000} - 1 = 140.42\dots$. We can take $N = 141$.
- (b) For the second series, we arrive at the inequality $(N+1)^{N+1} > 20000$. There is no general method to solve it. However, we can check different values of N . We have $5^5 = 3125$, but $6^6 = 46656$. Thus, we take $N+1 = 6$, i.e. $N = 5$.

The root test

Problem 28 Prove the Root Test:

Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers such that the limit

$$L = \lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}$$

exists. Then the following holds.

- (1) If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.
- (2) If $L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.
- (3) If $L = 1$, then the test is inconclusive.

Solution.

- (1) Suppose that $L < 1$. Take $r \in \mathbb{R}$ such that $L < r < 1$. By the definition of the limit with $\varepsilon = r - L > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n|^{\frac{1}{n}} < r \quad \text{for all } n \geq N.$$

This implies that

$$|a_n| < r^n \quad \text{for all } n \geq N.$$

Since the geometric series $\sum_{n=1}^{\infty} r^n$ converges, by the Comparison Test the series $\sum_{n=1}^{\infty} |a_n|$ converges, i.e. the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.

- (2) Now assume that $L > 1$. Using again the definition of the limit with $\varepsilon = L - 1 > 0$, we conclude that there exists $N \in \mathbb{N}$ such that

$$|a_n|^{\frac{1}{n}} > 1 \quad \text{for all } n \geq N.$$

But then $|a_n| > 1$ for all $n \geq N$. In particular, (a_n) does not converge to 0. Thus, the series $\sum_{n=1}^{\infty} a_n$ diverges.

- (3) The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent with

$$L = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln(\frac{1}{n})} = e^0 = 1.$$

The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is conditionally convergent with the same L .

Finally, the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is absolutely convergent with

$$L = \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} \right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} e^{\frac{1}{n^2} \ln(\frac{1}{n^2})} = e^0 = 1.$$